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பொறிக்களம்

 **DCKAP**
INCUBATION CENTER

STRUC^{IR}IRON

BUILD SMART. STAND STRONG.

RULEBOOK



REGISTRATION FEE: Rs. 149 /- PER TEAM

TEAM SIZE: MAX OF 3 PERSONS PER TEAM

PRIZE POOL MONEY: Rs. 2250 /-

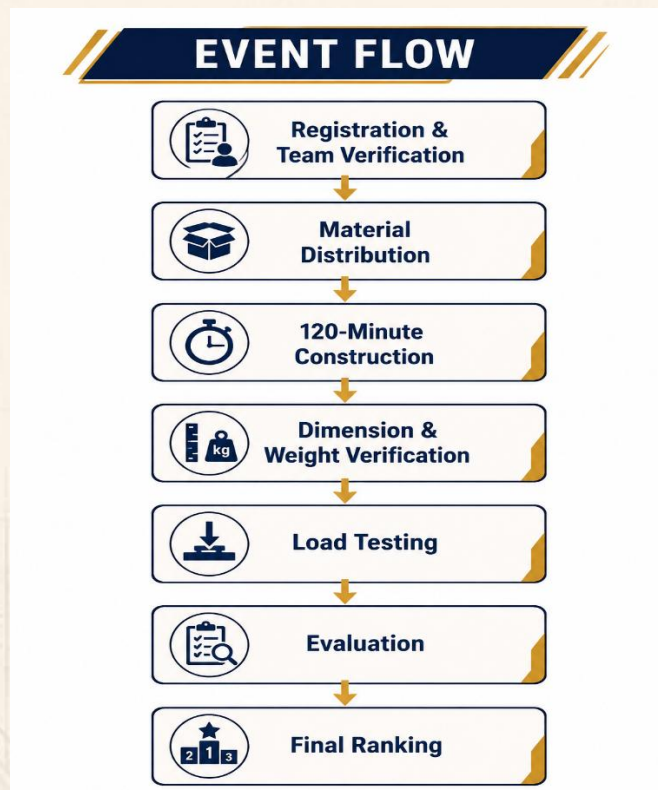
❖ Event Description

Design Smart. Analyze Strong.

Structron challenges you to put your structural thinking to work. Take on design and analysis challenges where every decision matters, every solution is tested, and engineering principles meet practical problem-solving. Think beyond formulas, approach the challenge strategically and develop structural solutions that stand strong.

❖ Event Flow

Design with precision. Analyze with purpose.





❖ *Team & Construction*

- Maximum 3 members per team.
- Each team can submit one tower only.
- Construction time: 120 minutes.
- All construction must be completed within the allotted time. No external assistance is permitted.
- Participants must carry their institute ID cards.

❖ *Tower Specifications*

Parameter	Requirement
HEIGHT	30 – 45 cm
BASE	NO FIXED LIMIT UNLESS ANNOUNCED
STRUCTURE	FREE - STANDING
MATERIALS	ORGANIZER - PROVIDED MATERIALS
SUBMISSION	ONE TOWER PER TEAM
TEST ATTEMPTS	ONE ATTEMPT PER TEAM

The tower must not be taped, nailed, glued, or fixed to the floor or any external support. Outside materials, prefabricated structures and kits are prohibited.



❖ *Materials Provided*

Construction materials will be issued at the beginning of the build period. These may include wooden sticks and other materials announced by the organizers. No additional materials will be issued after construction begins.

❖ *Load Testing*

After construction, each tower will undergo a controlled load test on the designated testing platform.

- Tower self-weight is measured before testing.
- Load is added incrementally through the loading platform.
- Testing continues until the tower fails.
- Collapse, tipping or significant structural damage constitutes failure.
- The maximum load sustained before failure is recorded.
- Each team receives one load-test attempt.



❖ Evaluation

Criterion	Marks	Evaluation
Maximum Load Sustained	60	Ability to withstand the highest load
Structural Efficiency	30	Load capacity relative to tower self-weight
Appearance & Design	10	Neatness, symmetry, creativity and workmanship
Total	100	

Structural Efficiency = Maximum Load Sustained ÷ Tower Self-Weight. A higher value indicates better structural performance for the tower's own weight.

❖ Penalties & Disqualification

A team may be penalized or disqualified for:

- Using unauthorized or outside materials.
- Exceeding the construction time.
- Violating the specified height.
- Modifying or touching the tower after testing begins.
- Interfering with another team's structure or testing.
- Using external assistance.
- Unsafe practices or intentional rule violations.



❖ *Tie-Breaker*

In case of an equal final score, ranking will be decided in the following order:

1. Higher Maximum Load Sustained
2. Higher Structural Efficiency
3. Higher Appearance & Design Score

If the scores remain equal after all three criteria, the decision of the organizers will be final and binding.

❖ *Final Authority*

The organizers may clarify or modify rules when required, with any changes communicated to the participating teams. The decision of the organizers.

FOR FURTHER DETAILS CONTACT:

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